

A.8 Raw Data Sources

We assessed the performance of FINCON using multi-modal financial data from January 3, 2022, to June 10, 2022, sourced from reputable databases and APIs including Yahoo Finance (via yfinance), Alpaca News API, and Capital IQ, detailed explained in Table. These data, initially stored in the Raw Financial Data Warehouse as available observations of the financial market environment, are diverged into the corresponding FINCON’s Analysts’ Procedural Memory Databases based on timeliness through working memory’s summarization operation.

Data Sources
News data associated with ticker: News data is sourced from REFINITIV REAL-TIME NEWS mainly contains news from Reuters.
Form 10-Q, Part 1 Item 2 (Management’s Discussion and Analysis of Financial Condition and Results of Operations): Quarterly reports (Form 10-Q) are required by the U.S. Securities and Exchange Commission (SEC).
Form 10-k, Section 7 (Management’s Discussion and Analysis of Financial Condition and Results of Operations): Annual reports (Form 10-K) are required by the U.S. Securities and Exchange Commission (SEC), sourced from EDGAR, and downloaded via SEC API.
Historical stock price: Daily open price, high price, close price, adjusted close price, and volume data from Yahoo Finance.
Zacks Equity Research: Zacks Rank: The Zacks Rank is a short-term rating system that is most effective over the one- to three-month holding horizon. The underlying driver for the quantitatively determined Zacks Rank is the same as the Zacks Recommendation and reflects trends in earnings estimate revisions. Zacks Analyst: Reason to Sell, Reason to Buy, and potential risks.
Earning Conference Calls (ECC): ECC is a type of unstructured financial data (audio) that is crucial for understanding market dynamics and investor sentiment. The company executive board delivers ECC about recent financial outcomes, future projections, and strategic directions. Recent studies have underscored the importance of not only the textual content of these calls but also the audio feature. Analyses have revealed that the audio elements—such as tone, pace, and inflections—offer significant predictive value regarding company performance and stock movements [94, 95, 96].

Table 6: Raw data and memory warehouses of FINCON

A.9 Distribution of Data

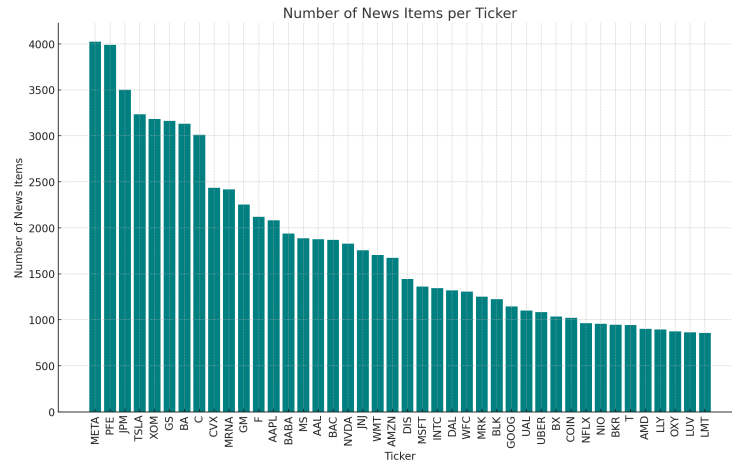


Figure 9: The distribution of news from REFINITIV REAL-TIME NEWS for the 42 stocks in the experiments

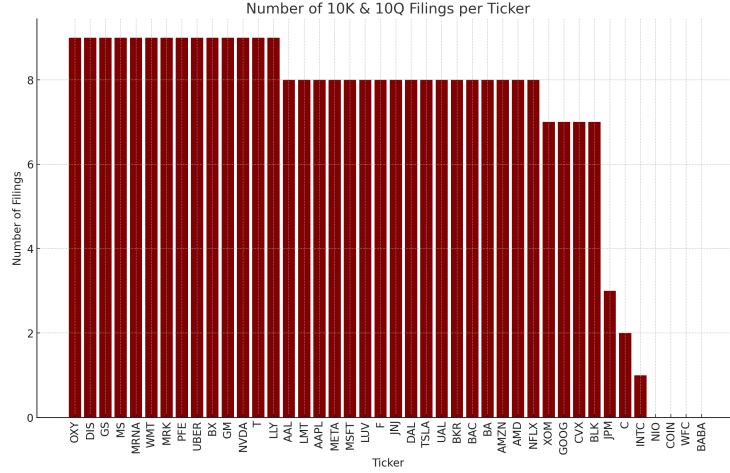


Figure 10: The distribution of 10k10q from Securities and Exchange Commission (SEC) for the 42 stocks in the experiments

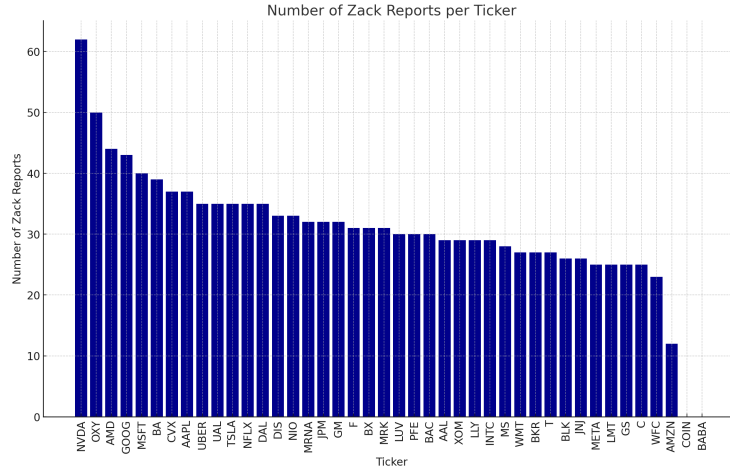


Figure 11: The distribution of Analyst Report from Zacks Equity Research for the 42 stocks in the experiments

A.10 Formulas of Classic Financial Metrics for Risk Estimator and Decision-making Task Performance Evaluation

The risk estimator uses the following metrics:

Profit and Loss (PnL)[97]: PnL quantifies the net outcome of trading activities over a specified period by accounting for the realized gains and losses from financial instruments like stocks and derivatives.

Value at Risk (VaR) of PnL[97]: VaR is a statistical tool used to estimate the potential loss in a portfolio, within a defined confidence interval. Mathematically, it is defined as Equation 3:

$$\text{VaR}_\alpha(PnL) = \inf \{l \in \mathbb{R} : \mathbb{P}(PnL \leq l) \geq \alpha\} \quad (3)$$

where α is the confidence level.

Conditional Value at Risk (CVaR) of PnL[97]: CVaR is a statistical tool used to estimate the expected potential loss worse than the VaR value in a portfolio, within a defined confidence interval. Mathematically, it is defined as Equation 4:

$$\text{CVaR}_\alpha(PnL) = \mathbb{E}\{PnL | PnL \leq \text{VaR}_\alpha(PnL)\} \quad (4)$$

where α is the confidence level.

The performance evaluation of algorithmic trading agents incorporates the following metrics:

Cumulative Return of PnL [90]: Cumulative Return is a key trading performance metric because it provides a comprehensive insight into investment performance, especially for strategies that emphasize long-term growth and reinvestment. The effectiveness of different investment strategies is evaluated based on their Cumulative Returns, which reflect the total change in value over time. In this study, we compute Cumulative Returns over the specified period by summing daily logarithmic returns, as outlined in Equation 5. This method is widely accepted in the finance area due to its ability to precisely capture minor price fluctuations and symmetrically address gains and losses. In essence, a higher Cumulative Return typically indicates a more effective strategy.

$$\begin{aligned} \text{Cumulative Return} &= \sum_{t=1}^n r_i \\ &= \sum_{t=1}^n \left[\ln \left(\frac{p_{t+1}}{p_t} \right) \cdot \text{action}_t \right], \end{aligned} \quad (5)$$

where r_i represents the PnL for day $t + 1$, p_t is the closing price on day t , p_{t+1} is the closing price on day $t + 1$, and action_t denotes the trading decision made by the model for that day.

Portfolio Value: Portfolio value represents the total worth of all the investments held in a portfolio at a given point in time. It is a metric used only in the portfolio management task.

$$\text{Cumulative Simple Return}_t = \prod_{k=1}^t (1 + \text{Daily Simple Return}_k) - 1 \quad (6)$$

$$\text{Portfolio Value}_t = \text{Initial Investment Amount} \times (1 + \text{Cumulative Simple Return}_t) \quad (7)$$

, where the initial amount is set as \$1,000,000.

Sharpe Ratio of PnL [91]: Sharpe Ratio is another core metric for evaluating investment performance and adjusting returns for risk. It is calculated by dividing the portfolio's average PnL (R_p) over the risk-free rate (R_f) by its volatility (σ_p), as shown in Equation 8. This metric adjusts returns for risk, with a higher ratio indicating better risk-adjusted performance. Essential in comparing different portfolios or strategies, it contextualizes performance against similar investments. Although a Sharpe Ratio above 1 is typically considered favorable and above 2 as excellent, these benchmarks can vary depending on the context of comparison.

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p} \quad (8)$$

Max Drawdown of PnL [92]: Max Drawdown is a metric for assessing risk. It represents the most significant decrease in a portfolio's value, from its highest (P_{peak}) to its lowest point (P_{trough}) until a new peak emerges, detailed in Equation 9. Indicative of investment strategy robustness, a smaller Max Drawdown suggests reduced risk.

$$\text{Max Drawdown} = \max \left(\frac{P_{\text{peak}} - P_{\text{trough}}}{P_{\text{peak}}} \right) \quad (9)$$

A.11 Baseline and Comparative Models on Single Stock Trading Task

Buy-and-Hold strategy (B&H):

A passive investment approach, where an investor purchases stocks and holds onto them for an extended period regardless of market fluctuations, is commonly used as a baseline for comparison of stock trading strategies.